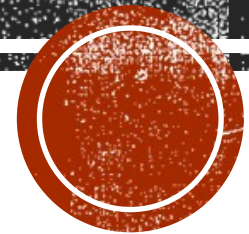


TUBERCULOSIS



Dr. Birhanu Kassie Reta, MD

Definition

- Infection is usually caused by the bacteria called **Mycobacterium Tuberculosis**



1. **Mycobacterium tuberculosis:**

- Cause pulmonary tuberculosis.
- Transmission is direct, by **inhalation of airborne** organisms in aerosols by exposure to contaminated secretions of infected persons.

2. **Mycobacterium bovis :**

- Cause oropharyngeal and intestinal tuberculosis.
- Transmitted by drinking **contaminated milk**.



3. Mycobacterium avium -intracellulare

- Less virulent than M. tuberculosis and **rarely** cause disease in immunocompetent individuals.



A. Primary Tuberculosis

- Primary tuberculosis is the form of disease that develops in a previously unexposed, **unsensitized**, person.
- The source of the organism is **exogenous**.
- Primary TB is generally asymptomatic, but leads to a positive PPD



- The inhaled bacilli implant in the distal airspaces of the **lower part of the upper lobe** or the **upper part of the lower lobe**, usually close to the pleura.
- 1- to 1.5-cm area of gray-white inflammatory consolidation formed, the **Ghon focus**.
- In most cases the center of this focus undergoes **caseous necrosis**.
- The combination of parenchymal lesion and nodal involvement is referred to as the **Ghon complex**.



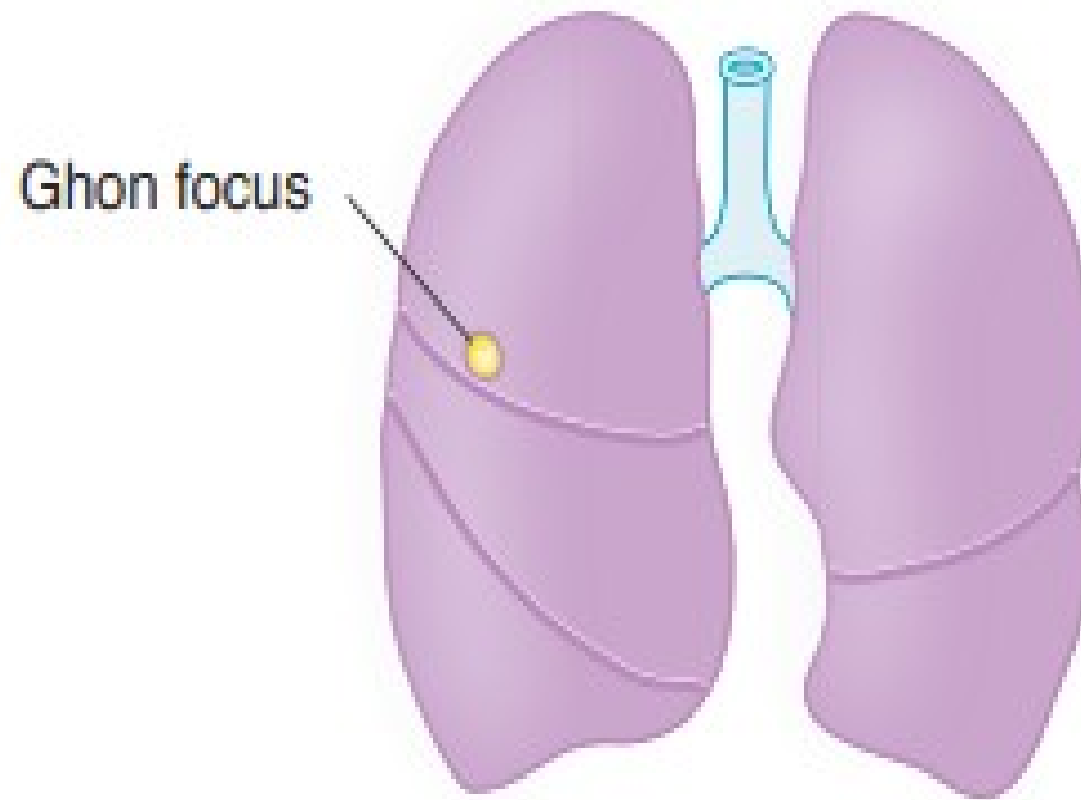


Figure 14-1. Primary Tuberculosis



The primary tuberculosis either:

(1) Form foci of scarring that harbor viable bacilli for years, and thus be the nidus for reactivation at a later time when host defenses are compromised.

or

(2) Progress to progressive primary tuberculosis (develop in individuals who are immunocompromised e.g, AIDS).



B. Progressive primary tuberculosis

- The apical lesion enlarges with expansion of the area of caseation.
- Erosion into a bronchus evacuates the caseous center, creating **irregular cavity lined by caseous material** that is poorly walled off by fibrous tissue.
- Erosion of blood vessels results in **hemoptysis**. With adequate treatment, the process may be arrested, although healing by fibrosis often distorts the pulmonary architecture.



- Additionally, dissemination to other organ systems can occur in advanced TB via a hematogenous route that often results in a miliary pattern within each affected organ.
- Sites that may become involved include meninges; cervical lymph nodes (scrofula) and larynx; liver/spleen, kidneys, adrenals, and ileum; lumbar vertebrae bone marrow (Pott disease); and fallopian tubes and epididymis

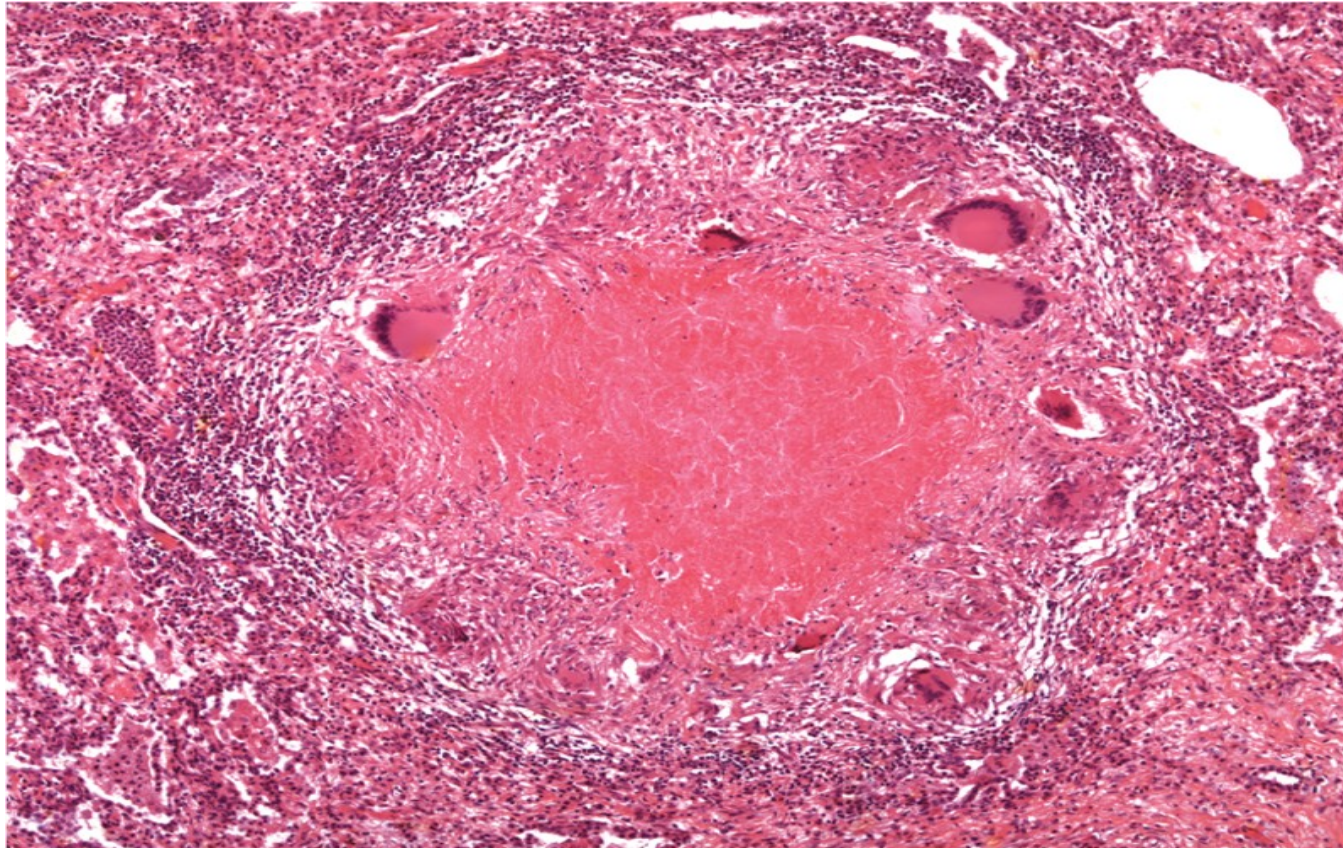


C. Secondary pulmonary TB

- Secondary (or post primary) tuberculosis is the pattern of disease that arises in a **previously sensitized** host.
- Arises from **reactivation of dormant primary lesions** many years after initial infection, particularly **when host resistance is weakened**.
- Secondary pulmonary tuberculosis is classically localized to the **apex of one or both upper lobes** (due to high oxygen tension in the apices).
- The **regional lymph nodes** are less prominently involved early in the developing disease than they are in primary tuberculosis.



CASEATING GRANULOMA



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- Sites that may become involved include meninges; cervical lymph nodes (scrofula) and larynx; liver/spleen, kidneys, adrenals, and ileum; lumbar vertebrae bone marrow (Pott disease); and fallopian tubes and epididymis



➤ Bone TB

- When the vertebrae are affected, the disease is referred to as **Pott disease**.
- **Paraspinal abscesses** in persons with this disorder may track along the tissue planes to present as an abdominal or pelvic mass.



POTT DISEASE



PARASPINAL ABSCESSES



➤ Lymph node TB

- Usually occurring in the cervical region ("scrofula").



➤ Intestinal TB

- Often a complication secondary tuberculosis, secondary to the **swallowing of coughed-up infective material**.
- Commonly affected site is the **ileum**.



➤ **TB of Central nervous system**

- TB Meningitis
- Tuberculoma
- TB arachdonitis



Clinical presentation

- Chronic persistent productive cough
- Systemic symptoms, related to cytokines released by activated macrophages (e.g., TNF and IL-1), include:
 - ✓ Malaise
 - ✓ Anorexia
 - ✓ Weight loss
 - ✓ Fever
 - ✓ Night sweats



- ✓ Increasing amounts of sputum, at first mucoid and purulent later. Hemoptysis is present in about half of all cases of pulmonary tuberculosis.
- ✓ Pleuritic pain may result from extension of the infection to the pleural surfaces.



- ✓ Extrapulmonary manifestations of tuberculosis are depend on the organ system involved (e.g, tuberculous salpingitis may present as infertility, tuberculous meningitis with headache and neurologic deficits, Pott disease with paraplegia).



➤ **Diagnosis**

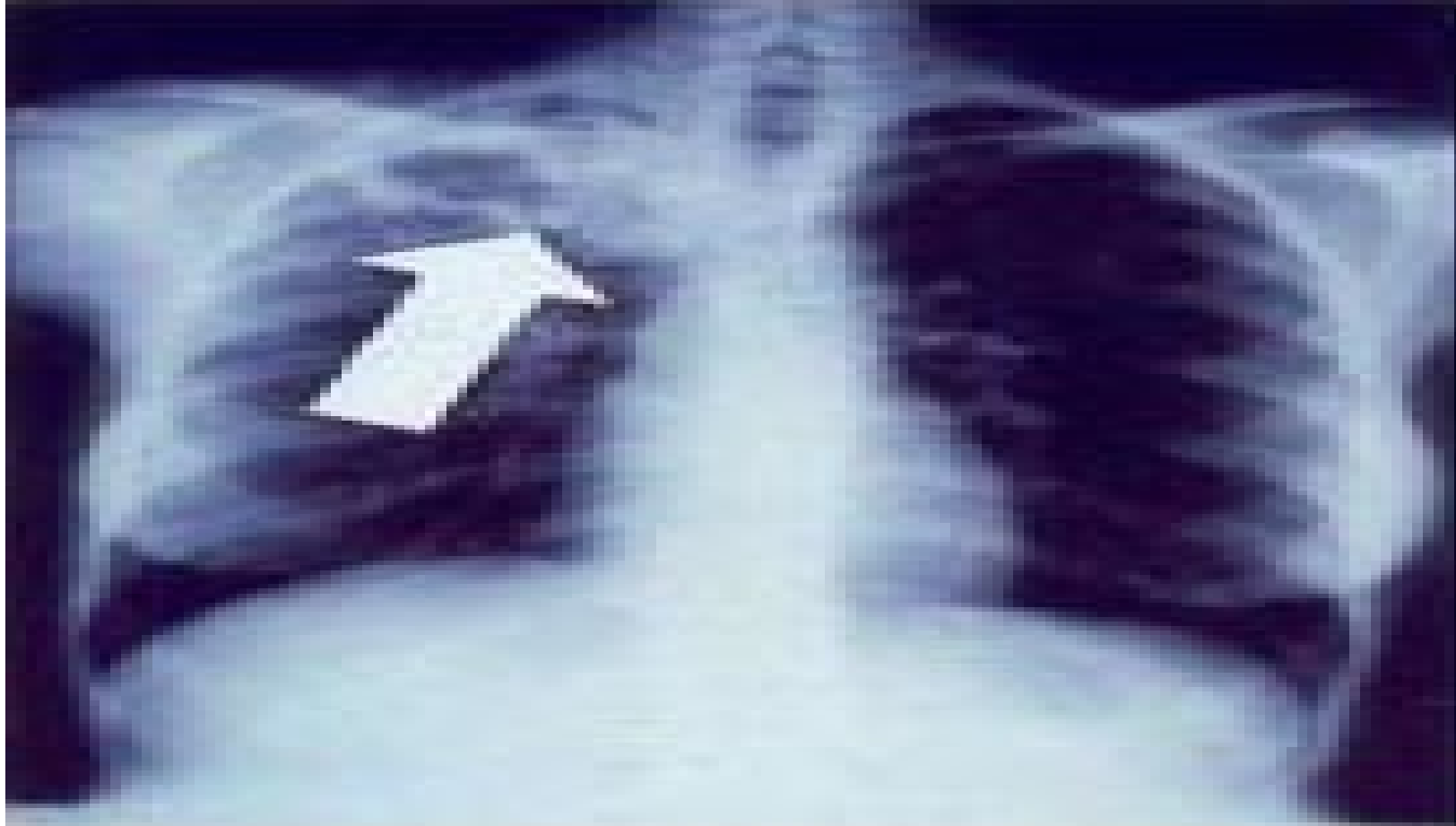
1. Demonstration of acid-fast organisms in sputum by acid-fast stains .
2. Conventional cultures for mycobacteria require up to 10 weeks.
3. PCR amplification of *M. tuberculosis* DNA.



4. Mantoux test (Tuberculin test): Tuberculin is an extract of *Mycobacterium tuberculosis*, *M. bovis*, or *M. avium* (purified protein derivative (PPD)) is used in skin testing to identify a tuberculosis infection.



CXR



➤ Vaccination for TB

- BCG (Bacillus Calmette-Guérin) is a tuberculosis vaccine prepared from a strain of attenuated live bovine tuberculosis bacillus, *Mycobacterium bovis*



Thank you!

